# rePhlow Minimum Viable Product: Technical Specification

## Purpose

This document holds the engineering detail behind the Minimum Viable Product section of the rePhlow wiki. The wiki page states what we build and how we will know it worked. This document states the specification of each component and the reasoning behind each choice, for judges or reviewers who want to verify that the headline claims are supported.

Nothing here is new relative to the wiki page. It is the same MVP at higher resolution.

The validation roadmap, the regulatory route and the techno-economic assessment are deliberately absent. They belong to the Business Development Plan section, and are listed at the end of this document as a handover.

## Scope

The MVP is a single functional unit: one five litre bench reactor with a rotating permeable basket, loaded with capsules of fixed formulation, operated in batch on real degumming effluent, coupled to a test ultrafiltration cartridge, plus a separate bench that processes the extracted polyphosphate.

It exists to place three hypotheses at risk.

1. **H1, technical.** The surface enzymatic layer releases orthophosphate from the organic phosphorus of a real degumming effluent, at pH around 5 and 30 ºC, at a rate compatible with an industrially reasonable retention time.
2. **H2, biological.** The synthetic PAO chassis accumulates that orthophosphate as polyphosphate inside the capsule, under non sterile conditions and in the real effluent matrix, without loss of containment.
3. **H3, value.** The extracted polyphosphate is a usable phosphoryl donor for a biocatalytic cycle, demonstrated here through ATP regeneration with BcPPK2-III and DHAP synthesis.

H3 sits inside the MVP rather than being deferred. If H1 and H2 succeed but H3 fails, rePhlow remains a phosphorus removal technology and loses its value proposition entirely.

| Deferred from the MVP | Why |
|---|---|
| Continuous operation at plant flow rate | Kinetics must be characterised in batch before a design HRT can be set |
| Multiple modules in parallel | Parallelisation is a scale up hypothesis, not a viability hypothesis |
| Full edit set (Δppx, ΔppkB, ΔpitB) | The MVP uses the minimum sufficient subset |
| Complete four family enzyme cocktail | The MVP tests the dominant route of this effluent, not generality |

| Deferred from the MVP | Why |
|---|---|
| Downstream cascade coupled in line | Coupling introduces confounding variables into the analytical proof |
| Cost optimisation, TEA, LCA | These consume performance data the MVP has yet to generate |

## Hardware

A 5 L rotating bed reactor with a permeable basket, coupled to a tangential flow ultrafiltration cartridge. Sampling ports on reactor, permeate and retentate, because the phosphorus balance only closes with all three.

| Parameter | Specification |
|---|---|
| Total and working volume | 5 L total, 3.0 to 3.5 L working |
| Basket | Cylindrical, 316L stainless steel mesh, aperture 0.8 to 1.0 mm, volume 0.8 to 1.2 L |
| Basket filling | 50 to 60 % v/v, capsules plus inert beads |
| Inert beads | Borosilicate glass or PTFE, 4 to 6 mm, inert to capsule ratio 1:2 to 1:3 v/v |
| Rotation | 10 to 120 rpm, PID controlled, continuously logged |
| Temperature | 30 ± 1 ºC, jacketed |
| pH | In line probe, no active correction unless drift exceeds 0.5 units |
| UF module | Hollow fibre, 100 kDa or 0.02 µm, 0.05 to 0.1 $m^2$, PVDF or PES, tangential flow |

Three of these choices carry the argument. The mesh aperture sits at least a factor of two below capsule diameter, so deformation cannot produce escape. The rpm range is wide on purpose: the deliverable is the curve of rotation speed against mass transfer and capsule integrity, not a single operating point. And pH is left uncorrected, because operating at the native conditions of the effluent is part of the hypothesis rather than an inconvenience to be engineered away.

Alongside the KPIs, the hardware must deliver a residence time distribution measured with an inert tracer, and a phosphorus balance closing at 90 % or better in every run. Without a closed balance there is no publishable datum.

## Cellular biocatalyst

The MVP strain is *Pseudomonas putida* KT2440 carrying three modifications: **pstSCAB+ / ppk1+ / Δppx**. Two of the five planned edits are deferred.

| Modification | Vector | In the MVP | Reason |
|---|---|---|---|
| pstSCAB, high affinity transport | pSEVA631-pEM7 | Yes | Governs uptake at low residual |

| Modification | Vector | In the MVP | Reason |
|---|---|---|---|
| | | | concentration. Without it the discharge limit is unreachable |
| ppk1, polyphosphate synthesis | pSEVA2513 | Yes | The metabolic sink. Without it phosphorus enters but is not stored |
| Δppx, exopolyphosphatase | Chromosomal, markerless, pMBEC base editing | Yes | Direct antagonist of ppk1. With ppx active, net accumulation is dominated by a futile cycle |
| ΔppkB, consumption | Chromosomal | No | Second order effect on net balance. Optimisation, not viability |
| ΔpitB, efflux | Chromosomal | No | Relevant to maximising final yield, not to demonstrating accumulation |

The reason for stopping at three is risk management rather than effort. Each additional chromosomal edit adds construction time and one more confounding variable to the phenotype, and the MVP has to be able to attribute an accumulation failure to an identifiable cause.

**Entry gate.** Before encapsulation, the strain must show polyP accumulation statistically above wild type KT2440 in planktonic culture on defined medium, quantified by DAPI-polyP or toluidine blue, confirmed by exopolyphosphatase digestion and malachite green assay. Encapsulating an uncharacterised strain makes any negative result uninterpretable.

**Computational support.** The COBRA genome scale and COPASI kinetic models are not a parallel deliverable. Within the MVP they predict the growth against accumulation split under real Bio-Oils profiles, design the C:P ratio and HRT sweep, and are then contrasted against the experimental data. The quality of that fit is itself a KPI.

## Capsule

One formulation, fixed for the whole MVP campaign, so that any variation in performance is attributable to process conditions rather than to the carrier.

| Parameter | Specification |
|---|---|
| Core | Liquid cell suspension in compatible buffer, inoculum density recorded as OD600 or g DCW per L of core |
| Alginate | 2.0 % w/v, high G content, M/G characterised per batch |
| Gelation | 100 mM CaCl2, fixed curing time |
| Surface functionalisation | Three hydrolases immobilised directly on the alginate surface, see Enzymatic layer |
| Diameter | 2.5 ± 0.3 mm, monodispersity CV below 10 % |

A liquid core preserves viability and diffusion relative to a solid matrix. The M/G ratio governs stiffness and shear resistance, and therefore service life, which is why it is documented per batch rather than assumed. It also governs the surface presented for enzyme immobilisation, so the capsule formulation and the enzymatic layer cannot be optimised independently of each other.

**Per batch characterisation.** Diameter and distribution, compression resistance, and a magnetite nanoparticle containment test before and after mechanical cycling. The nanoparticles are strictly smaller than the bacterium, which makes the test conservative by construction.

**Screening required before real effluent work.** Resistance to chelation. The real effluent may contain species that sequester Ca2+ and destabilise the alginate network, and this is the most plausible route by which a capsule that performs well in synthetic medium fails on the real stream.

## Enzymatic layer

Three enzymes, not four. The MVP cocktail covers the dominant route of the degumming effluent rather than the full spectrum of organic phosphorus chemistry.

| Enzyme | In the MVP | Role |
|---|---|---|
| Phospholipase A, Lecitase Ultra | Yes, core of the MVP | The majority fraction of organic P in degumming is phospholipidic. Commercial, characterised, pH and temperature window matching the effluent. This is the enzyme that validates H1 |
| Acid phytase | Yes | Covers phytate, the second refractory fraction. Naturally acidic optimum |
| Non specific acid phosphatase | Yes, closing enzyme | Hydrolyses the monoesters released upstream through to free orthophosphate. |

| Enzyme | In the MVP | Role |
|---|---|---|
| | | Without it the cascade halts at intermediates pstSCAB cannot take up |
| Phospholipase C | No | Overlapping routes with PLA on the same substrate. Its incremental value is quantified once the PLA baseline exists |

**Immobilisation.** The three enzymes are immobilised directly on the alginate surface of the capsule. The MVP adapts the protocol already validated on a model support, that is, prior ionic adsorption followed by covalent cross linking, detergent free, to preserve orientation and active site. Transferring that protocol from an agarose based support to a calcium alginate surface is itself a risk to be closed early, since both the surface chemistry and the mechanical fragility of the support differ.

**Critical gate.** Two numbers must be reported. Specific activity retained after immobilisation, as U per g capsule against U per mg free enzyme. And enzyme leaching into the medium, as supernatant activity at t = 0, at 24 h, and after each cycle.

The second number decides whether the concept holds. An enzymatic layer that leaches is not an enzymatic layer: it is soluble enzyme dosing with extra steps, and it removes the reusability argument on which the economics rest.

## Downstream valorisation bench

The recovered polyphosphate is a phosphoryl donor rather than a finished product, and what it becomes depends on the enzymes it is fed to. The cascade specified below is therefore a proof of concept rather than a commitment to a product: it establishes that rePhlow derived polyphosphate can drive a biocatalytic cycle, and the same donor can feed other cascades whose output is selected according to industrial demand, environmental benefit or economic return.

That cascade is run decoupled from the bioreactor. Direct coupling would introduce matrix inhibition, phosphatase carryover and ambiguity as to the origin of the phosphate, and the objective at this stage is an unambiguous analytical proof rather than an integrated process.

1. **Capsule recovery** from the basket at the end of the accumulation cycle.
2. **Controlled lysis and polyP extraction** under a fixed protocol, with extraction yield determined against an internal control.
3. **Characterisation of the extracted polyP**: quantification by DAPI-polyP and malachite green after hydrolysis, plus chain length distribution by urea PAGE with toluidine staining. Chain length is not a detail. BcPPK2-III has substrate preference, so a polyP that is quantitatively abundant may still be functionally poor.
4. **ATP regeneration**: BcPPK2-III with extracted polyP and AMP or ADP. Quantified by luciferase assay, confirmed by HPLC or LC-MS of the nucleotide pool. Two independent methods, because luciferase is sensitive to matrix interferents.

5. **Demonstrator cascade**: BcPPK2-III with PROSS stabilised DHAK and dihydroxyacetone, yielding DHAP, with polyP as the sole energetic phosphate source and nucleotide present in catalytic rather than stoichiometric amount. Quantified by coupled GPDH/NADH assay and by LC-MS.

**Controls.** Four, and they are what makes the result publishable rather than suggestive.

- Commercial polyP of comparable chain length, as positive reference
- Reaction without polyP
- Reaction with free orthophosphate in place of polyP
- Capsule extract from a non accumulating wild type strain

The last control is the one that demonstrates the phosphorus originates from engineered accumulation rather than from carryover.

**The metric that defines value rather than function.** Nucleotide total turnover number, meaning how many phosphorylation cycles each AMP or ADP molecule supports before loss of activity. This figure sustains the claim of major cost saving against stoichiometric ATP. Without it, that claim is rhetorical.

## Validation KPIs

These are proposed decision thresholds, to be calibrated with Bio-Oils and with the first characterisation data. They are not performance promises, and we recommend presenting them on that basis.

| KPI | Definition | Minimum | Target | Method |
|---|---|---|---|---|
| Organic P conversion | P-PO4 released / initial organic P, real effluent, native pH, 30 ºC | 50 % in 8 h | 70 % in 6 h | Total P after digestion against soluble reactive P |
| Capture yield | P removed / initial soluble P, per batch | 60 % | 80 %, residual below discharge limit | Balance on UF permeate |
| Specific capacity | mg P accumulated per g DCW | 30 | 60 | Intracellular polyP quantification |
| Capture kinetics | HRT required to reach capture yield | 12 h or less | 6 h or less | Depletion curve, serial sampling |
| Containment, cells | CFU in UF permeate | 0, LOD below 1 CFU per 100 mL | 0, all cycles | Plate count on concentrated volume |
| Containment, DNA | Strain specific chromosomal marker in | Below LOQ | Below LOQ | qPCR on a unique target |

| KPI | Definition | Minimum | Target | Method |
|---|---|---|---|---|
| | permeate | | | |
| Capsule integrity | Intact capsules after N cycles | 95 % after 5 | 95 % after 10 | Counting plus post cycling magnetite test |
| Enzymatic stability | Residual hydrolytic activity after N cycles | 70 % after 5 | 70 % after 10 | Standardised activity assay |
| Biological stability | Capture yield at cycle N against cycle 1 | 70 % after 5 | 70 % after 10 | Balance repetition |
| UF stability | TMP increase at constant flux per cycle | Below 15 %, backwash recoverable | Below 10 % | TMP logging |
| ATP regeneration | Conversion using extracted polyP against commercial reference | 60 % of reference | 85 % | Luciferase plus LC-MS |
| Nucleotide TTN | Phosphorylation cycles per nucleotide molecule | 20 | 100 | DHAP produced / nucleotide added |
| DHAP synthesis | Unambiguous production from rePhlow derived polyP | Two orthogonal methods, four controls | 70 % conversion on DHA | GPDH/NADH plus LC-MS |
| Model validation | Deviation between predicted and observed capture | Below 30 % | Below 15 % | Prediction against experiment |

The last KPI is worth defending explicitly, because it is the one a reader is most likely to dismiss as academic. A model that predicts system behaviour within a known margin is a scale up asset: it allows the number of modules for a new client to be sized without repeating a full pilot campaign. That turns computational modelling into a reduction in commercial deployment cost.

## Handover to the Business Development Plan

Four blocks of material were developed alongside this MVP but belong to the next wiki section. They are listed here so the content is not lost, and so the two sections do not repeat each other.

**Validation roadmap beyond the bench.** Phase 2 moves to real unmodified effluent with multi round cycling and full downstream execution. Phase 3 installs a single module in bypass on a minor fraction of the Bio-Oils Huelva flow, traversing the natural variability of the degumming process: oil batch changes, shutdowns, restarts. Gate for progression to scale up: four or more weeks of continuous operation within thresholds and no containment incidents.

**Real matrix failure modes to screen.** Chelation of alginate Ca2+, fouling by fat content, enzymatic inhibition, microbial competition under non sterile conditions.

**Regulatory route.** Phase 3 involves operating a GMO outside a research facility. Under Directive 2009/41/EC on the contained use of genetically modified microorganisms and its Spanish transposition, this requires risk classification of the activity, notification to the competent authority, and accreditation of the bypass installation as a contained use space. The file should be opened early rather than at the point of need, since administrative timelines are typically the real bottleneck in the schedule.

**Techno-economic and life cycle assessment.** Both are Phase 3 deliverables, because both consume operating data the MVP generates. A cost figure per kg of phosphorus recovered, published before that data exists, weakens the claims around it that are properly supported.